Chiranjeev Singh

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SKILLS

C/C++ | Python | Java | Fast-API | Javascript | Node.js | Typescript | SQL | Pytorch | Algorithms | Data Structures | Flask | Version Control | Machine Learning | NLP | Data Modelling | Agentic AI | Multimodal RAG | AWS | Azure | GCP | Guice | Docker | API Design | Bash | LLM | Langchain | Docker | Kafka | Redis | System Design | Agentic workflow orchestration

PROFESSIONAL EXPERIENCE

SDE (L60), *Microsoft* — 04/2026 – Present, Bengaluru

- Built an anomaly detection system for Blips & Pauses (BP), improving detection of **unexplained IO latency anomalies by ~20%** using large-scale telemetry and unsupervised ML. Subsequently, developed a **release comparison framework** to evaluate two releases across **IO latency, workload, and network metrics**, enabling more efficient performance analysis, regression identification, and **operational follow-ups.**
- Contributed to **overprovisioning** and capacity throttling mechanisms, **improving resource utilization efficiency by ~15%** under bursty IO workloads.
- Developed and validated Per-Disk Policy for automatic disk recovery on **IO timeouts (E17),** significantly **reducing VM downtime scenarios (~80%)** in failure testing.
-Worked on distributed storage architecture involving slice-based replication and **Block Storage Services (BSS)**, strengthening **system durability and reliability.**

SDE - JWO Tech, *Amazon* — 05/2025 – 02/2026, Bengaluru, India

- **Devised** and executed a **runtime resource allocation service** that **replaced hardcoded configurations** with runtime **S3-based caching**, improving flexibility and maintainability across **10+ downstream services** with **100%** daemon **uptime.**
- Implemented **Dynamic Size Limiting** for IMOS, **reducing SEV2** tickets by approximately **30%** and **improving Inquiry Dwell Time** by up to **23 minutes per rejected request**, while eliminating manual queue management interventions.
- Successfully migrated a **legacy grant management** and creation service from **Angular, CoffeeScript, and Ruby build system** to **React + Polaris** with **100% uptime**, reducing operational engineering load by approximately **15%** and mitigating associated **SAS risks**.
- Increased **service success rate from 85% to ~92%** by introducing a **timeout-aware spillover mechanism**, efficiently discarding delayed requests and **improvin**g overall **system efficiency**.
- **Led** internal AI/ML knowledge sessions on **Neural Networks/Transformers and GenAI**, attended by **200+ employees** across orgs, followed by interactive **Q&A engagements**.

Oyo

SDE 1 — 07/2024 – 04/2025, Gurugram, India

- **Led cross-functional development** of a data pipeline using **Azure ACS** and **Cognitive Services** for a **Call Center Automation bot**, driving a **projected 20% cost reduction** for Belvilla and Dancenter.
- Built a **voice processing pipeline** for a performance dashboard, **achieving 86% sentiment accuracy** on **1,000+ clips** and **improving issue resolution by 20%**.
- Improved **digital audit efficiency** by **optimizing and migrating pipelines**, boosting performance and **reducing latency**.
- **Optimized retrieval-augmented workflows** for chatbots across multiple brands, **increasing search speed by 15%** and **handling 2,000+ daily queries** per department.

SDE Intern — 01/2024 – 07/2024, Gurugram, India

- Engineered s**patial clustering pipelines** using **DBSCAN** and **HDBSCAN** algorithms to analyze user and hotel location data, identifying fraudulent bookings with a **4:1 True/False positive ratio**.
- Developed and optimized **GenAI model pipeline** for hero image selection, incorporating **advanced masking techniques** which contributed to a **20% increase in hotel booking conversion rates** by refining visual content.

EDUCATION

Bachelors of Engineering, *Thapar Institute of Engineering and Technology* — 10/2020 – 07/2024, Patiala, India

- Major in Computer Science and Engineering (9.30 CGPA).
- **Coursework** : Machine learning (AI/ML), Deep learning, Conversational AI ,Object Oriented Programming (OOPS), Database management system (DBMS), Operating System (OS), Computer Networks (CN)

CERTIFICATES

- Microsoft Certified: Azure Data Scientist Associate
- AI Programming with Python - Nanodegree
- Microsoft Certified - Azure AI Fundamentals (AI-900)
- Machine Learning Fundamentals - Nanodegree

ACHIEVEMENTS AND FELLOWSHIPS

Merit Scholarship, *Thapar Institute of Engineering and Technology* — 2024

- Merit scholarship for **Outstanding Academic Perfomance** 4 years in a row, for the terms of 2020-2024.

AWS AI/ML Scholar, Winter Cohort, *Udacity* — 2022

- Among the 500 people worldwide that were awarded the Scholarship and proceeded to **Phase 2 - Advanced ML Nanodegree**.